

Puzzles to Puzzle You (Remastered)

Welcome to the third iteration of *Puzzles to Puzzle You (Remastered)* by Rani Rachavelpula, in which I will discuss the previous puzzle (Dido's Problem) and pose the next one.

I want to begin by saying that I do not believe in apologies (If you are sorry, then you shouldn't have done it). Therefore, I will not apologize for the tardiness of this month's newsletter.

Nevertheless, I do believe in reparations. In accordance with this steadfast belief, I made myself learn two proofs of Dido's Problem, one utilizing differential geometry and another utilizing Fourier Analysis. The latter proof was particularly painful. Through this gesture, I hope my commitment to this newsletter and, most importantly, to this community is no longer in question.

Puzzle #2 (Dido's Problem)

1. Determine from all simple closed planar curves of the same perimeter p the one that encloses maximal area.

Circle

2. Prove its maximality (bonus).



Queen Dido founding Carthage (1657) from Rare Book Division, The New York Public Library.

The solution to this puzzle was known by Dido, and she used it to maximize the area of the ancient city Carthage.

There is something unsettling about Dido's problem. It is easy to see the answer is a circle, but it is difficult to give a good reason why it must be a circle. To give a rigorous proof is even more difficult. In fact, in the 19th century, Weierstrass noted that no one had a good reason for believing a solution to Dido's problem exists given that similar problems were shown to lack solutions. Weierstrass himself was only able to give a proof of Dido's problem after having invented a new branch of mathematics called calculus of variations which deals with optimizing functionals, the class of functions to which an integral belongs.

Since Weierstrass, mathematicians have come up with numerous other proofs of Dido's problem. In what follows I will provide two such proofs, one utilizing differential geometry and another utilizing Fourier Analysis. Before turning to these proofs, some shout outs.

Shout out (Andrew) My dear friend Andrew reviewed his notes on the calculus of variations to give a proof of Dido's problem. Such dedicated readership is deeply moving. We love you Andrew!

Shout out (Harry) I always knew my friend Harry, a skilled woodworker, possessed great geometrical insight. One night after dinner at a local barbecue restaurant, Harry surprised me with an elegant and simple argument for Dido's problem. First he gave two reasons why the optimal curve must be smooth (no corners) and convex. Then he explained that the global optimality of the curve depends on its local optimality at any restricted region. This implies that the curve is maximally symmetric. (An n -manifold is *maximally symmetric* iff it possesses the largest possible number of linearly independent Killing vectors $\frac{n(n+1)}{2}$.) This implies it must be a circle.

Maybe: Harry

Ok my proof

H

Once you get to all concave and no corners

Edited

It can't be an egg because obviously the shape has to be symmetrical cause whatever is optimal area for one half has to be the other half

But that has to be true for all forms of symmetry and ad infinitum the only shape that is perfectly symmetrical is circle

H

QED

Harry's original proof

If that wasn't enough, Harry latter observed that since bubbles minimize surface tension, bubbles are the solution to Dido's problem in 3D. Though the extra dimension complicates some things, Harry's insight about bubbles is generally correct. After all, only a fool would argue with nature!

Proof (Dido's Problem)

To prove Dido's Problem, I'll prove the Isoperimetric Inequality.

Theorem (Isoperimetric Inequality): Let Γ be a simple, closed planar curve of length l . Then,

$$A \leq \frac{l^2}{4\pi}$$

Corollary (Dido's Problem): Equality holds when Γ is a circle.

Proof 1 (Differential Geometry)

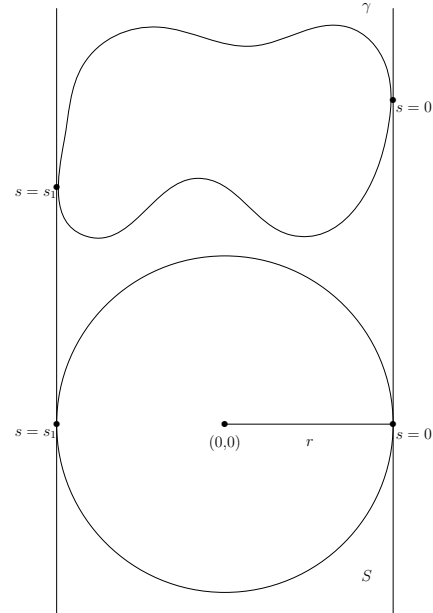
STEP 1: Define γ to be a parametrization of Γ by arc-length that is positively oriented. Then,

$$\begin{aligned}\alpha_\gamma(s) &= (x_\gamma(s), y_\gamma(s)), s \in [0, l] \\ |\alpha'_\gamma(t)| &= \sqrt{(x'_\gamma(s))^2 + (y'_\gamma(s))^2} \\ &= 1\end{aligned}$$

STEP 2: Since γ is closed, we can bound γ by vertical lines with two points of tangency as shown below. Draw a circle S below γ , also

with two points of tangency. Impose a coordinate system such that the origin is the center of the circle. Parametrize the circle such that

$$\begin{aligned}\alpha_S(s) &= (x_S(s), y_S(s)) \\ &= (x_\gamma(s), y_\gamma(s))\end{aligned}$$



STEP 3: Recall, the formula for the area of a closed planar curve from calculus

$$Area(A) = \int_0^l x(s)y'(s)ds$$

By the product rule,

$$\int_0^l (xy)'ds = \int_0^l xy'ds + \int_0^l x'yds$$

But since γ closed

$$\begin{aligned}\int_0^l (xy)'ds &= xy(1) - xy(0) \\ &= 0\end{aligned}$$

$$\Rightarrow \int_0^l xy'ds = - \int_0^l x'yds$$

Using this, write

$$\begin{aligned}A_\gamma &= \int_0^l x_\gamma y'_\gamma ds \\ A_S &= - \int_0^l x'_S y_S ds \\ &= - \int_0^l x'_\gamma y_S ds \\ &= \pi r^2\end{aligned}$$

STEP 4: Compute $A_\gamma + \pi r^2$.

$$\begin{aligned}
 A_\gamma + \pi r^2 &= A_\gamma + A_S \\
 &= \int_0^l x_\gamma y'_\gamma ds - \int_0^l x'_\gamma y_S ds \\
 &\leq \int_0^l \sqrt{(x_\gamma y'_\gamma - x'_\gamma y_S)^2} ds \\
 &\leq \int_0^l \sqrt{(x_\gamma + y_S)^2 (x'_\gamma + y'_\gamma)^2} ds \\
 &= \int_0^l \sqrt{(x_\gamma + y_S)^2} ds \\
 &= \int_0^l \sqrt{(x_S + y_S)^2} ds
 \end{aligned}$$

The penultimate step follows since γ is parametrized by arc length. Thus,

$$\begin{aligned}
 A_\gamma + \pi r^2 &= \int_0^l \sqrt{(x_S)^2 + (y_S)^2} ds \\
 &= \int_0^l r ds \\
 &= lr
 \end{aligned}$$

STEP 5: Notice that

$$\begin{aligned}
 \sqrt{A}\sqrt{\pi r^2} &\leq \frac{1}{2}(A + \pi r^2) \\
 &\leq \frac{1}{2}lr \\
 &\Rightarrow A \leq \frac{l^2}{4\pi}
 \end{aligned}$$

This completes the proof of the Isoperimetric Inequality. $\square$

STEP 6: Consider the case of equality. Let

$$\begin{aligned}
 A_\gamma &= \frac{l^2}{4\pi} \\
 &= lr - \pi r^2
 \end{aligned}$$

Therefore

$$\begin{aligned}
 4\pi lr - 4\pi^2 r^2 - l^2 &= 0 \\
 \Rightarrow (2\pi r - l)^2 &= 0 \\
 \Rightarrow l &= 2\pi r, A_\gamma = \frac{(2\pi r)^2}{4\pi} = \pi r^2
 \end{aligned}$$

This shows the solution to Dido's problem is a circle. $\square$

Proof 2 (Fourier Analysis)

This proof is particularly illuminating. It re-describes a simple, closed curve as a 2π -periodic function to which we can apply the powerful techniques of Fourier Analysis.

STEP 1: Notice that the Isoperimetric Inequality is invariant under re-scalings. That is, we are free to rescale the length l . WLOG, let $l = 2\pi$. This reduces our proof to the following statement:

$$A \leq \pi$$

STEP 2: Define γ to be a parametrization of Γ by arc-length that is positively oriented. This means that $x'(s)^2 + y'(s)^2 = 1$, which implies that

$$\frac{1}{2\pi} \int_0^{2\pi} x'(s)^2 + y'(s)^2 ds = 1$$

STEP 3: Since our curve is closed, both x, y are 2π -periodic. Consider their Fourier series

$$\begin{aligned}
 x &\sim \sum_{n=-\infty}^{\infty} a_n e^{ins} \\
 y &\sim \sum_{n=-\infty}^{\infty} b_n e^{ins}
 \end{aligned}$$

This implies,

$$\begin{aligned}
 x' &\sim \sum_{n=-\infty}^{\infty} (a_n in) e^{ins} \\
 y' &\sim \sum_{n=-\infty}^{\infty} (b_n in) e^{ins}
 \end{aligned}$$

STEP 4: By Parseval's Identity, we have

$$\sum_{n=-\infty}^{\infty} |a_n in|^2 = \frac{1}{2\pi} \int_0^{2\pi} |x'(s)|^2 ds$$

Thus,

$$\begin{aligned}
 \sum_{n=-\infty}^{\infty} |a_n in|^2 + |b_n in|^2 &= \sum_{n=-\infty}^{\infty} (n^2)(|a_n|^2 + |b_n|^2) \\
 &= \frac{1}{2\pi} \int_0^{2\pi} |x'(s)|^2 + |y'(s)|^2 ds \\
 &= 1
 \end{aligned}$$

STEP 5: This next step involves two formulas. The first is the area formula utilized in the previous proof. After some algebraic manipulation, we have

$$A = \frac{1}{2} \left| \int_0^l x(s)y'(s) - x'(s)y(s) ds \right|$$

The second formula is the bilinear form of Parseval's Identity, which states the following:

If $F \sim \sum a_n e^{inx}$, $G \sim \sum b_n e^{inx}$, then

$$\frac{1}{2\pi} \int_0^{2\pi} F \overline{G} dx = \sum_{n=-\infty}^{\infty} a_n \overline{b_n}$$

Since x, y real valued $a_n = \overline{a_{-n}}$. Putting the two formulas together we, have

$$\begin{aligned} A &= \frac{\pi}{2} \left| \int_0^l x(s)y'(s) - x'(s)y(s) ds \right| \\ &= \pi \sum_{n=-\infty}^{\infty} a_n \overline{b_n} n - \overline{a_n} n b_n \\ &= \pi n \sum_{n=-\infty}^{\infty} a_n \overline{b_n} - \overline{a_n} b_n \end{aligned}$$

STEP 6: Bound $|a_n \overline{b_n} - \overline{a_n} b_n|$, using the triangle inequality and Cauchy-Schwartz

$$\begin{aligned} |a_n \overline{b_n} - \overline{a_n} b_n| &\leq 2|a_n||b_n| \\ &\leq |a_n|^2 + |b_n|^2 \end{aligned}$$

STEP 7: Finally, use this to bound our expression for the area

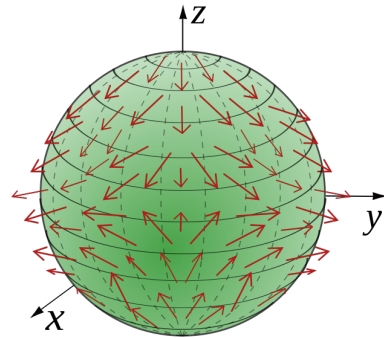
$$\begin{aligned} A &= \pi n \left(\sum_{n=-\infty}^{\infty} a_n \overline{b_n} n - \overline{a_n} n b_n \right) \\ &\leq \pi n^2 \left(\sum_{n=-\infty}^{\infty} a_n \overline{b_n} n - \overline{a_n} n b_n \right) \\ &\leq \pi n \left(\sum_{n=-\infty}^{\infty} |a_n|^2 + |b_n|^2 \right) \\ &\leq \pi \end{aligned}$$

This completes the second proof of the Isoperimetric Inequality $\square$.

Puzzle #3: 4D Hedgehog Theorem

Imagine a smooth wind on the surface of the Earth always parallel to the ground. It famously follows from the Hedgehog Theorem (also known as the Hairy-Ball Theorem) that the speed of the wind (represented below by arrows) must vanish for at least one point on the Earth's surface.

1. Does this result hold for 4 dimensions?
If the surface of the Earth was a 3-sphere embedded in 4 dimensions, does it still hold that there must exist a point where the wind is speed 0?
2. Why (bonus)?



The 3D Hedgehog Theorem was first proved by Poincaré in 1885. This theorem has many useful applications. It explains why a coconut necessarily has a tuft and why you cannot comb a dog without admitting a bald spot.

The Hairy-Ball Theorem

Those are a few of the concepts and objects studied by topology: now we'll look at a theorem.

If you look at the way the hairs lie on a dog, you will find that they have a 'parting' down the dog's back, and another along the stomach. Now topologically a dog is a sphere (assuming it keeps its mouth shut and neglecting internal organs) because all we have to do is shrink its legs and fatten it up a bit (Figure 90).



Figure 90

One might wonder whether it is possible to comb the hairs in such a way that all partings were eliminated. This would give a smooth